

DIGITAL LENDING: PORTFOLIO OPTIMIZATION

A credit policy blueprint built from 39,913 loans - who to lend to, what to charge, and how to catch defaults two months before they happen.

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39,913 loans · ₹377.5 cr disbursed · 36 cohorts
4 findings → 1 recommendation → 1 impact number

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GROWTH IS STRONG. THE ECONOMICS ARE NOT.



Veritas Capital Finance (our fictional digital NBFC) lends across three products - personal loans, BNPL and SME working capital - to urban and semi-urban India, including first-time and underbanked borrowers.



Three years of rapid scaling built a ₹377.5 crore book. But the bottom line is thin: credit losses are eating almost everything the interest earns.



The CRO's brief: use end-to-end customer and transaction data to strengthen risk assessment, catch delinquency early, fix pricing and product design - and keep growing without degrading the book.

Our answer: it can be fixed - but only by accepting that some of the current growth is not worth having.

₹377.5 cr

Disbursed across 39,913 loans (Jan 2023 – Dec 2025)

13.7%

Of disbursed value has ever crossed 90+ days past due

₹22.8 cr

Gross credit losses by the Mar 2026 snapshot

1.42%

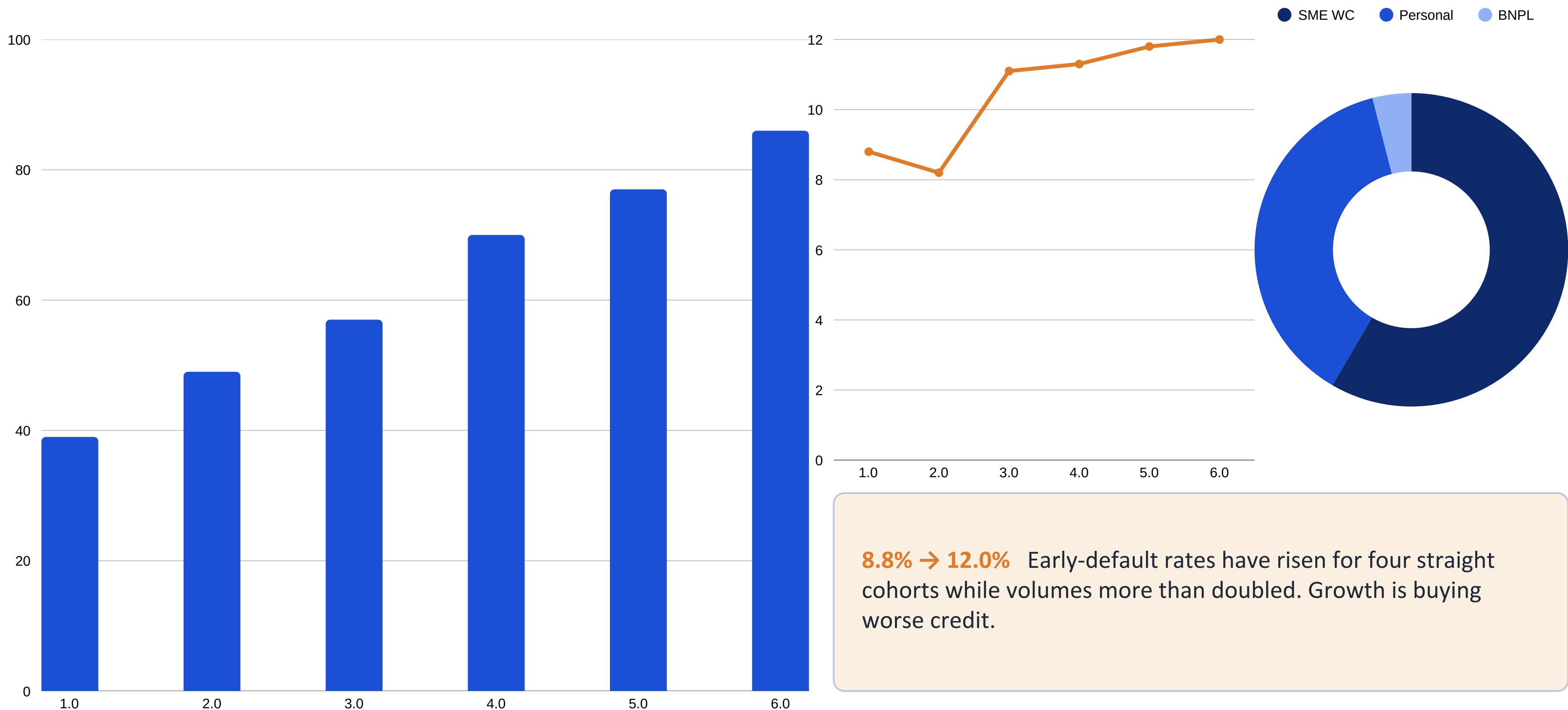
Risk-adjusted contribution margin — before cost of funds

WE BUILT THE EVIDENCE BASE THE BRIEF ASKED FOR

24,000 borrowers	39,913 loans	36 monthly cohorts	3,16,962 repayment records
 CUSTOMER PROFILE Geography (metro / tier-2 / tier-3), income proxy, employment type, bureau band incl. thin-file	 LOAN & PRODUCT Product, ticket size, tenure, APR, origination risk grade from a deliberately imperfect scorecard	 REPAYMENT BEHAVIOUR Monthly on-time / partial / 30-DPD / 90+ status for every loan	 BEHAVIOURAL SIGNALS Cash-flow consistency, balance volatility, spending shocks (bank-statement style)
 ACQUISITION Channel, cost of acquisition, approval turnaround time	 TIME DIMENSION Originations Jan 2023 – Dec 2025; performance snapshot Mar 2026	 OUTCOMES Default flag, credit loss, contribution- margin value per loan	 BUILT-IN REALISM The brief's causal relationships embedded under statistical noise — nothing visible without analysis

Reproducible end-to-end: seeded data generator + analysis code ship with this deck. Re-run it and you get these exact numbers.

EACH COHORT IS BIGGER — AND WORSE



Methodology: every cohort measured at the same age (6 months on book) — no seasoning artifact.

RISK IS CONCENTRATED — AND VISIBLE UPFRONT

Segments built from three things we know on day one: bureau band × employment type × cash-flow volatility

Segment	Share	Default	Margin	Avg APR
S1 Anchor Salaried Prime	18.6%	3.6%	+3.99%	19.9%
S2 Established Self-Employed	23.9%	8.8%	+5.46%	21.4%
S3 Stable New-to-Credit	17.0%	16.3%	−0.05%	23.1%
S4 Mid-Book Mixed	28.9%	16.5%	+0.80%	22.8%
S5 Volatile Gig & Informal	7.2%	24.5%	−5.65%	23.9%
S6 Stressed Subprime	4.4%	36.6%	−10.07%	24.9%

10×

Default spread, safest to riskiest segment — 3.6% vs 36.6%, knowable at application

11.6% → 24.5%

Share of book held by S5+S6 → share of all credit losses they generate

The franchise is NOT the safest segment.

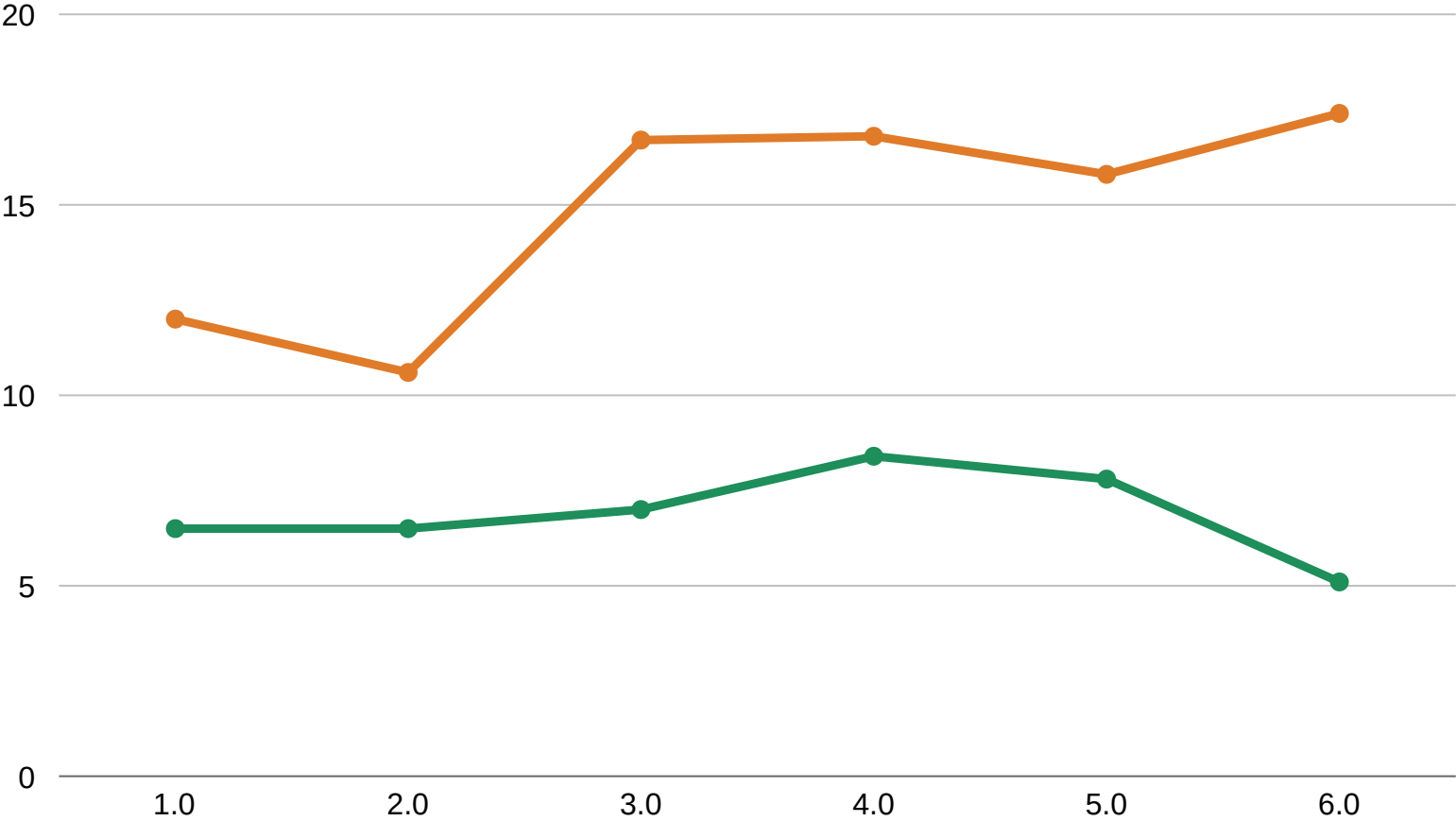
Established Self-Employed accepts 21.4% APR, defaults at a manageable 8.8%, and earns the book's best margin.

Risk is knowable on day one — the application file already contains it.

THE GROWTH ENGINE IS ADVERSELY SELECTED

Channel	Share of value	CAC / loan	Approval TAT	Default (value)	Margin
DSA / Aggregator	33.8%	₹3,196	2.6 h	19.0%	-2.7%
Digital Marketing	30.4%	₹2,303	6.0 h	12.9%	+2.1%
Organic / Repeat	26.0%	₹451	8.5 h	9.0%	+5.6%
Partnership / Embedded	9.9%	₹1,095	1.7 h	10.4%	+2.4%

● DSA / Aggregator ● Organic / Repeat



Aggregator share of new lending:
23% (2023) → 42% (2025)

Aggregator 6-month default runs
~2× organic in every cohort

Classic adverse selection: DSA partners are paid on volume, so applicants declined elsewhere flow to the fastest approver. Speed isn't the problem — Partnership is even faster and performs fine. Volume-incentivised middlemen are.

CLEAR SWEET SPOTS, CLEAR TRAPS

Product × ticket band	Share	Default	Margin	Verdict
SME WC — under ₹2.5L	12.0%	12.7%	+3.9%	Core engine — protect
SME WC — ₹2.5–6L	36.3%	16.0%	+3.2%	Core engine — protect
SME WC — above ₹6L	10.1%	19.7%	+1.9%	Cap — tail risk builds
Personal Loan — ₹60–150k	24.4%	10.6%	+1.6%	Sweet spot at 12-month tenure
Personal Loan — above ₹150k	6.2%	17.1%	−0.3%	Restrict to grade A/B repeats
BNPL — under ₹8k	1.1%	1.4%	−31.9%	Cost problem, not credit



TENURE COMPOUNDS RISK

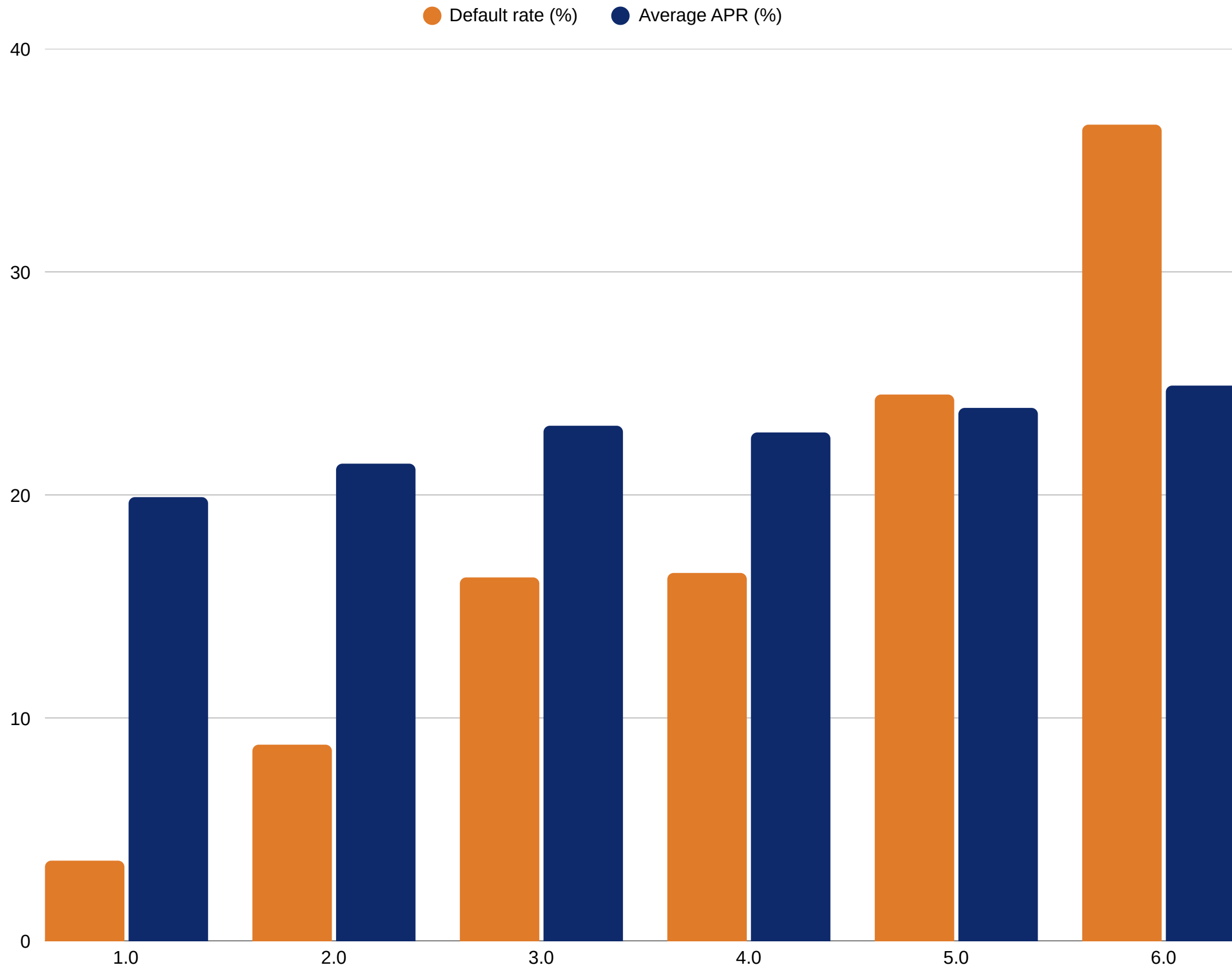
36-month personal loans default at 18.4% vs 8.9% at 12 months — margin flips from +1.7% to −1.4%. Long tenure means more time for life to go wrong.



BNPL IS A COST PROBLEM, NOT A CREDIT PROBLEM

Sub-₹8k BNPL defaults at just 1.4% yet loses ~32 paise per rupee: fixed CAC and servicing swamp tiny fee income. Fix = ticket floors + partner-funded acquisition — not tighter approvals.

PRICE MOVES 5 POINTS. RISK MOVES 33.



APR band: 19.9% → 24.9%
Default band: 3.6% → 36.6%

Best borrowers are OVER-charged relative to their risk — an open invitation for competitors to refinance them away.

The risky middle is UNDER-charged — every incremental rupee written there loses money.

Risk-based pricing exists in name — premia compressed far below losses.

DEFAULTS ANNOUNCE THEMSELVES ~2 MONTHS EARLY



TRIGGER 1 — PARTIAL-PAYMENT PATTERN

Partial payments in 2 of the last 3 months

77.2%

of eventual defaulters flagged before their default month

2 months

median lead time — 39.1% flagged 2+ months ahead



TRIGGER 2 — FIRST MISSED CYCLE

Any 30-DPD event, even if cured next month

4.0× lift

30.4% of flagged accounts default vs a 7.5% base rate

14.4%

of never-default accounts get flagged — the workload cost, stated honestly

Backtested on 3,16,962 monthly repayment records, 2023–2025.

FIRST FLAG → app nudge + payment-restructure offer

→ ABOVE ₹1L EXPOSURE → RM call within 7 days

→ UNTIL CURED → freeze top-ups; 2nd flag feeds origination rules

One default found per 3 calls on the flag list — instead of 1 per 13 from undirected calling.

A THREE-MOVE POLICY PACKAGE, SEQUENCED OVER TWO QUARTERS



1. EXIT THE RED ZONE

IMMEDIATE

Decline aggregator-sourced grade D/E applicants who are subprime OR cash-flow volatile.

- 8.5% of volume · 35.6% default · -9.2% margin · 22% of ALL losses
- Renegotiate DSA payouts to first-loss / trail commissions
- Cap the channel at 30% of monthly originations



2. REPRICE THE AMBER MIDDLE

NEXT PRICING CYCLE

+200 bps on C/D grades outside the red zone (~half the book); retention pricing for A/B grades.

- Cap personal loans at ₹150k / 24 months (except A/B repeats)
- ₹8,000 BNPL ticket floor outside partner-subsidised programmes
- Cap SME exposure above ₹6 lakh pending collateralised variants



3. STAND UP EARLY WARNING

60-90 DAYS

Deploy the two-trigger rule into collections with the tiered playbook.

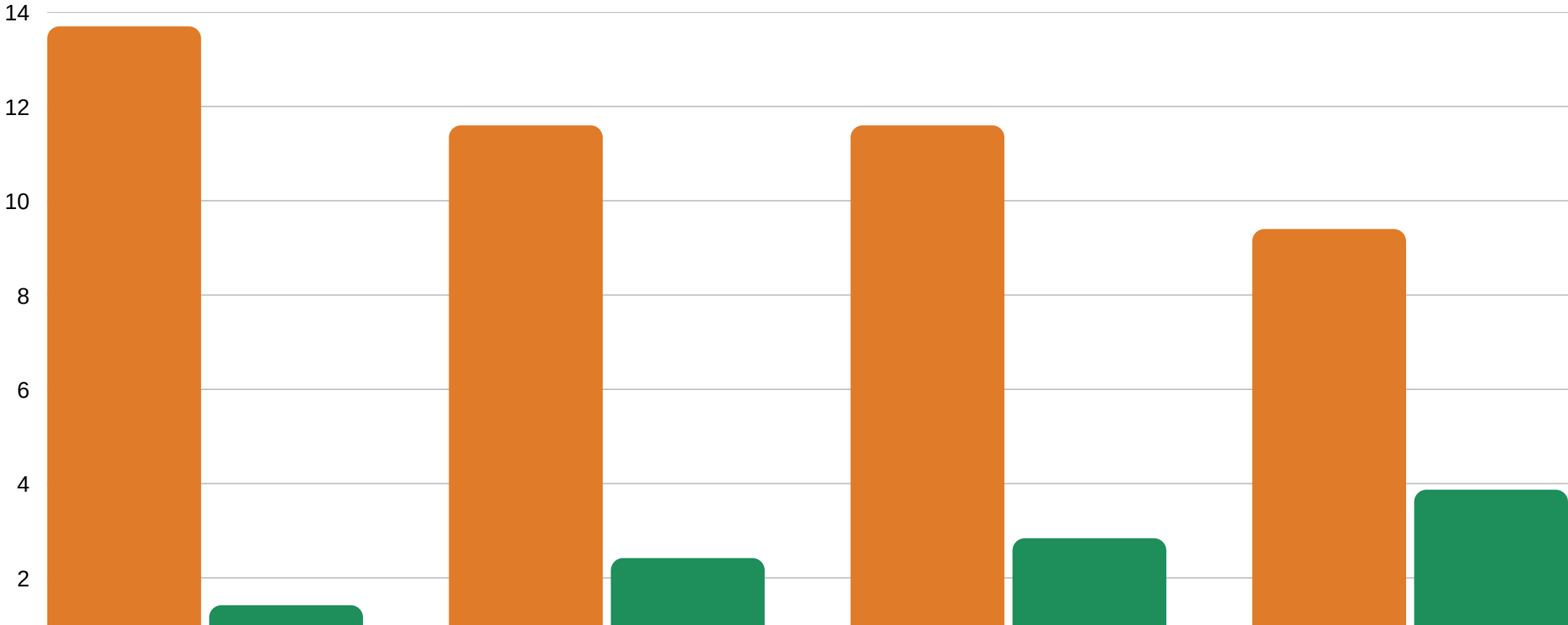
- Saves ~₹3.3 crore/year of losses at a conservative 25% cure rate
- Second flags feed origination as behavioural decline rules
- Cure rate measured from month one against the assumption

Each move stands alone — the next slide stacks their economics.

MARGIN NEARLY TRIPLES FOR 8.5% LESS VOLUME

Metric	Today	+ Red-zone exit	+ Repricing	+ Early warning
Default rate (% of value)	13.7%	11.6%	11.6%	9.4%
Contribution margin (%)	1.42%	2.42%	2.84%	3.87%
Origination volume retained	100%	91.5%	~84%	~84%

● Default rate (%) ● Margin (%)



≈ ₹9 crore / year

of additional risk-adjusted value on the current book once all three moves land

STATED ASSUMPTIONS

- LGD: 70% PL / 82% BNPL / 58% SME
- Average outstanding balance: 55% of ticket
- 15% volume attrition in response to +200 bps
- 25% cure rate on early-warning outreach
- Margins measured before cost of funds

Direction and order of magnitude, not a forecast — every assumption is on the table to be stress-tested.

WHAT COULD GO WRONG — AND WHAT WE DO ABOUT IT

RISK — Topline dip: ~8.5% volume lost now, up to ~16% with repricing attrition; growth optics suffer 2–3 quarters.

MITIGATION — Redirect acquisition spend to Organic (+5.6%) and Partnership (+2.4%); judge growth on margin per rupee, not volume.

RISK — Repricing adverse selection: if competitors hold prices, only our weakest applicants accept the +200 bps.

MITIGATION — Cap the increase at 200 bps; pair with retention pricing for A/B grades; review take-up mix monthly.

RISK — Financial inclusion optics: tightening can read as excluding thin-file borrowers.

MITIGATION — Red zone targets channel × grade × volatility — never thin-file status. S3 stays fully open.

RISK — Collections workload: EWS flags 14.4% of good accounts (~1 false flag per 7 good accounts/yr).

MITIGATION — Tiered response keeps cost low: app nudges first; human calls only above ₹1 lakh exposure.

RISK — Assumption risk: the 25% cure rate and 15% attrition are estimates, not facts.

MITIGATION — Measure both from month one; recalibrate the red zone on post-policy cohorts in Quarter 2.

We state the trade-offs before anyone asks — and the package is still clearly worth it.

LEADERSHIP DASHBOARD — AND THE FIRST 90 DAYS

THE MONTHLY ONE-PAGE RISK VIEW

- 1 **6-month vintage default by cohort × channel** — earliest clean read on origination quality
- 2 **EWS flag rate & flag→default conversion** — the health of the middle book
- 3 **Channel risk-adjusted margin incl. CAC** — growth judged on value, not volume
- 4 **Red-zone share of new originations** — policy adherence
- 5 **Roll rates: current→30 and 30→90 DPD** — collections effectiveness
- 6 **Top-decile exposure concentration** — SME tail risk

THE 90-DAY IMPLEMENTATION PATH

- 1 **DAYS 0–30**
Red-zone decline rule live in the loan origination system; DSA renegotiation opens; leadership dashboard v1 ships.
- 2 **DAYS 30–60**
EWS triggers wired into the collections CRM; tiered playbook training; repricing grid approved by ALCO.
- 3 **DAYS 60–90**
Repriced grid live; first post-policy vintage read-out; cure-rate measurement begins against the 25% assumption.
- 4 **QUARTER 2**
Recalibrate: tighten or relax the red-zone definition on observed data; extend EWS to top-up eligibility.

DATA SOURCES & REFERENCES

- Synthetic dataset designed per the problem statement's Data Design Guidelines — seven data areas, expected relationships embedded under statistical noise
- Generator & analysis: Python (NumPy, Pandas) — `generate_data.py`, `analyze.py`, `summary.json`; seeded and fully reproducible
- Every chart in this deck is plotted from `summary.json` — each figure traceable back to code
- Assumption benchmarks: unsecured-retail LGD ranges, pre-delinquency cure rates and repricing elasticity drawn from public industry literature (RBI Financial Stability Reports, TransUnion CIBIL industry insights, listed-NBFC disclosures)
- Economics convention: $\text{contribution margin} = \text{interest} + \text{fees} - \text{credit losses} - \text{CAC} - \text{servicing}$; before cost of funds and overhead

Dataset + code: < https://drive.google.com/file/d/1_nQDJK-Kwtme6BLjVu3K2Dbpr_fouym-/view?usp=sharing >

THANK YOU

Trade 8.5% of volume for nearly 3× the margin.

Every number in this deck is reproducible from the accompanying dataset and code — happy to stress-test any assumption.